

MINI PROJECT REPORT

Retail Sales Performance Analysis

Excel Data Preprocessing &
Power BI Interactive Dashboard

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1. Introduction

The Retail Sales Analysis mini project focuses on preparing, analyzing and presenting retail sales data in a clear and interactive form. The project was completed in two major phases. Phase 1 used Microsoft Excel for data preprocessing, validation, calculated fields and Pivot Table analysis. Phase 2 used Power BI to transform the prepared data into an interactive multi-page dashboard.

The main purpose of retail sales analysis is to understand sales performance, profitability, customer-related patterns and order behaviour. A dashboard makes these results easier to interpret because users can apply filters, move between analytical pages, switch between bookmarked views and drill from summarized information to detailed transactions.

The final Power BI report contains four named pages: Retail Sales Overview, Sales & Profit Analysis, Customer & Order Analysis and Order Details. This exceeds the minimum three-sheet requirement and provides a structured presentation of the analysis.

2. Objectives

- To clean and validate the retail sales dataset before analysis.
- To handle blank rows, missing values, inconsistent categorical values and invalid numerical values.
- To create useful calculated fields and Pivot Table summaries during the Excel preprocessing phase.
- To analyze sales, profit, customer and order-related performance using Power BI visualizations.
- To develop an interactive dashboard using KPIs, charts, slicers, navigation buttons, bookmarks and drill-through.
- To provide a clear and user-friendly report that supports exploration of retail sales performance.

3. Dataset Description

The Retail Sales dataset used in the project was first imported into Excel. The original dataset was preserved in a Raw Data sheet and a separate Cleaned Data sheet was prepared for analysis. The final cleaned dataset contains 4,200 records and 23 columns, including calculated fields.

Dataset Attribute	Description
Records	4,200 records in the final cleaned dataset
Columns	23 columns, including calculated fields
Raw Data	Original dataset preserved separately
Cleaned Data	Validated and transformed dataset used for analysis
Domain	Sales & E-commerce / Retail Sales

3.1 Important Fields

- order_id – identifies an order/transaction.
- order_date – date associated with the order.
- customer_id / customer_name – customer identification and name information.
- region / city – geographic information used for regional analysis.
- product_category / product_name – product-level information used for category and product analysis.
- quantity – quantity associated with the transaction.
- sales_amount – sales/revenue value used in sales analysis.
- profit – profit value used for profitability analysis.
- payment_method – method used for payment analysis.
- order_status – status such as Delivered, Shipped, Pending, Cancelled and Returned.
- gender, age and customer_satisfaction – customer-related fields used for demographic and satisfaction analysis.
- discount_amount and profit_margin – calculated fields created during preprocessing.

4. Tools & Technologies Used

Tool / Technology	Purpose
Microsoft Excel	Initial data preparation, cleaning, validation, filtering, sorting and Pivot Tables.
Power Query	Data transformation and preparation workflow where applicable.
Microsoft Power BI	Interactive dashboard development, visualization, filtering and navigation.
DAX	Measures/calculations used where applicable for KPI and analytical calculations.

5. Data Cleaning & Transformation

Data cleaning was completed in Phase 1 before the dataset was used for Power BI dashboard development. The process focused on improving data quality and ensuring that the fields were suitable for analysis.

Blank-row removal: Completely blank rows were identified and removed.

Missing values: Missing numerical values were checked and handled using a median-based approach where required.

Invalid age values: Age values were reviewed and invalid values were corrected.

Categorical standardization: Gender, Order Status and Payment Method values were checked and standardized.

Date standardization: Order Date was cleaned and converted into consistent Excel date values.

Numerical validation: Quantity, Unit Price, Discount %, Shipping Cost, Customer Satisfaction and Days to Ship were validated.

Duplicate checking: Duplicate records were checked and the cleaned dataset was validated to avoid retaining duplicate transactions.

Filtering and sorting: Excel filters and number filters were used to inspect ranges, missing values and unusual observations.

5.1 Calculated Fields

- discount_amount – created to represent the monetary discount associated with an order.
- profit_margin – created to support profitability analysis using Profit and Sales Amount.

5.2 Pivot Table Analysis

Pivot Table	Purpose
Region-wise Sales	Analyze sales performance by region.
Category-wise Sales	Compare sales contribution by product category.
Region-wise Profit	Compare profit performance by region.
Monthly Sales Trend	Analyze sales movement over time.
Order Status Summary	Summarize order-status distribution.

6. Data Analysis & Dashboard Development

Phase 2 transformed the cleaned retail sales data into an interactive Power BI dashboard. The dashboard was organized into four logical pages so that overall performance, sales and profit, customer and order analysis, and detailed transactions could be viewed separately.

6.1 Report Pages

Page	Sheet Name	Main Purpose
1	Retail Sales Overview	Overall sales performance
2	Sales & Profit Analysis	Regional, category and payment analysis
3	Customer & Order Analysis	Customer, gender, order and return analysis
4	Order Details	Detailed drill-through transaction view

6.2 Page 1 – Retail Sales Overview

This page acts as the main overview dashboard for quickly understanding overall retail sales performance.

- KPI cards: Total Sales, Total Profit, Total Orders and Total Quantity.
- Sales by Region bar chart.
- Sales by Product Category bar chart.
- Monthly Sales Trend line chart.
- Region, Product Category and Order Date slicers.
- Sales Analysis navigation button.

6.3 Page 2 – Sales & Profit Analysis

This page compares sales and profitability across regions, product categories and payment methods.

- Sales by Region visualization.
- Total Profit by Region visualization.
- Sales and Profit by Category visualization.
- Orders by Payment Method visualization.
- Overall Profit Margin card.
- Region, Product Category, Payment Method and Order Date slicers.
- Sales View and Profit View bookmarks.
- Sales View and Profit View buttons to switch between bookmarked views.
- Home navigation to the overview page.
- Customer Analysis navigation to the customer/order page.

6.4 Page 3 – Customer & Order Analysis

This page focuses on customer-related and order-related patterns using charts and interactive slicers.

- Customer Satisfaction by Region chart.
- Sales by Gender donut chart.
- Orders by Payment Method bar chart.
- Order Status Distribution donut chart.
- Return Analysis donut chart.
- Sales by Age visualization.
- Region, Gender, Payment Method and Order Status slicers.
- Home button and Sales Analysis button for navigation.

6.5 Page 4 – Order Details and Drill-through

The Order Details page provides a transaction-level view using Power BI drill-through. Region was configured as the drill-through field, allowing a selected region from an analytical page to open the detailed order view.

- Detailed transaction table.
- Order ID, Order Date, Customer Name, Region, City, Product Category and Product Name.
- Quantity, Sales Amount, Profit, Payment Method and Order Status.
- Region configured as the drill-through field.
- Back button to return from the drill-through page.

6.6 Interactive Features

Feature	Implementation
Buttons	Page navigation and bookmark switching buttons.
Bookmarks	Sales View and Profit View.
Drill-through	Region-based drill-through to Order Details.
Slicers	Region, category, date, gender, payment method and order status where applicable.
Navigation	Home, Sales Analysis, Customer Analysis and Back navigation.

6.7 Dashboard Design

The dashboard uses clear page names and headings to separate the analysis. Visuals are arranged to make KPI values, charts and slicers easy to identify. Consistent navigation and interactive controls improve usability and allow users to explore the report without returning to the report page tabs for every action.

7. Dashboard Screenshots & Visual Explanations

The completed Power BI dashboard contains four report pages. The following section documents the main visuals and explains what each page is intended to communicate. Screenshots of the final Power BI pages can be inserted below each subsection when preparing the final printed/submitted version.

Figure 1 – Retail Sales Overview

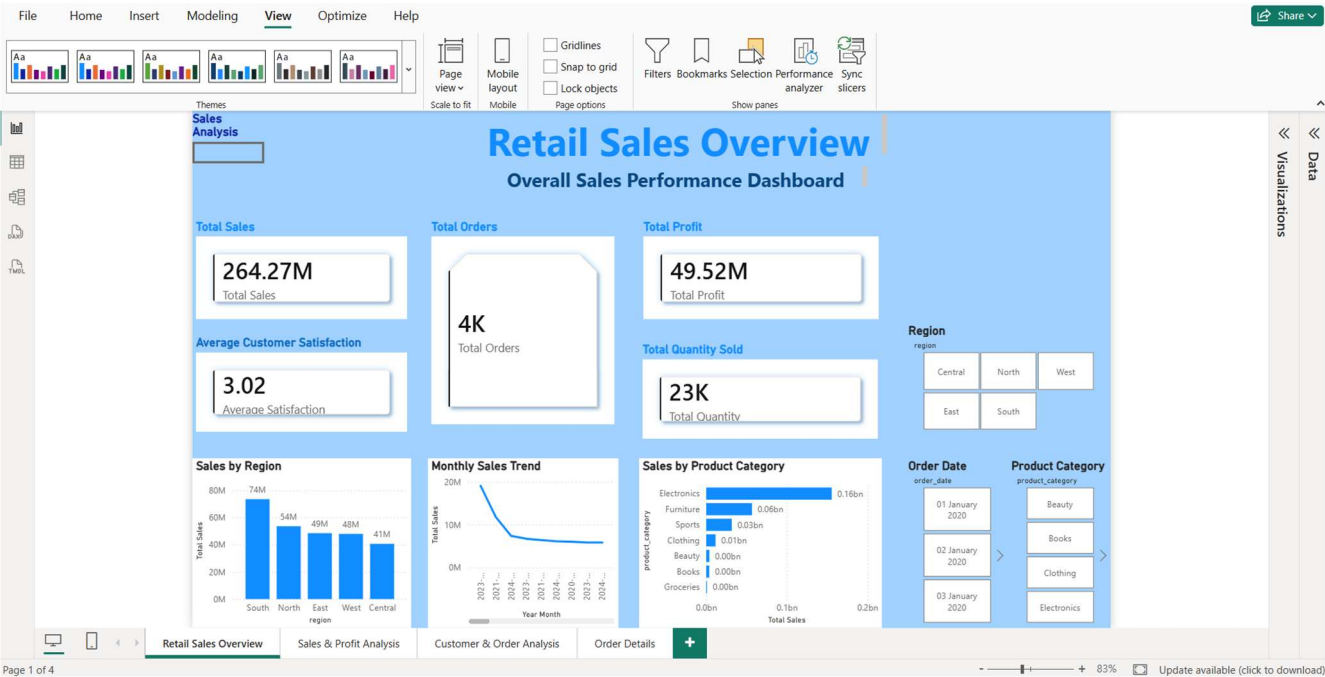


Figure 2 – Sales & Profit Analysis

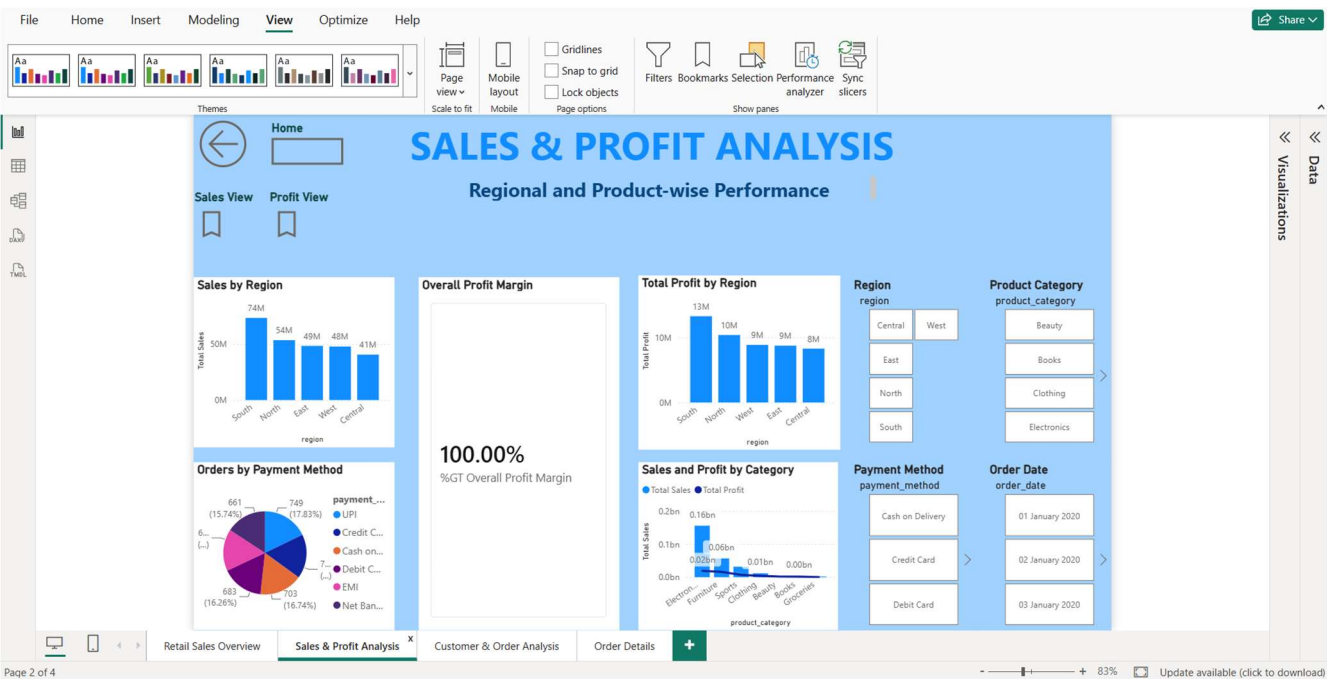


Figure 3 – Customer & Order Analysis

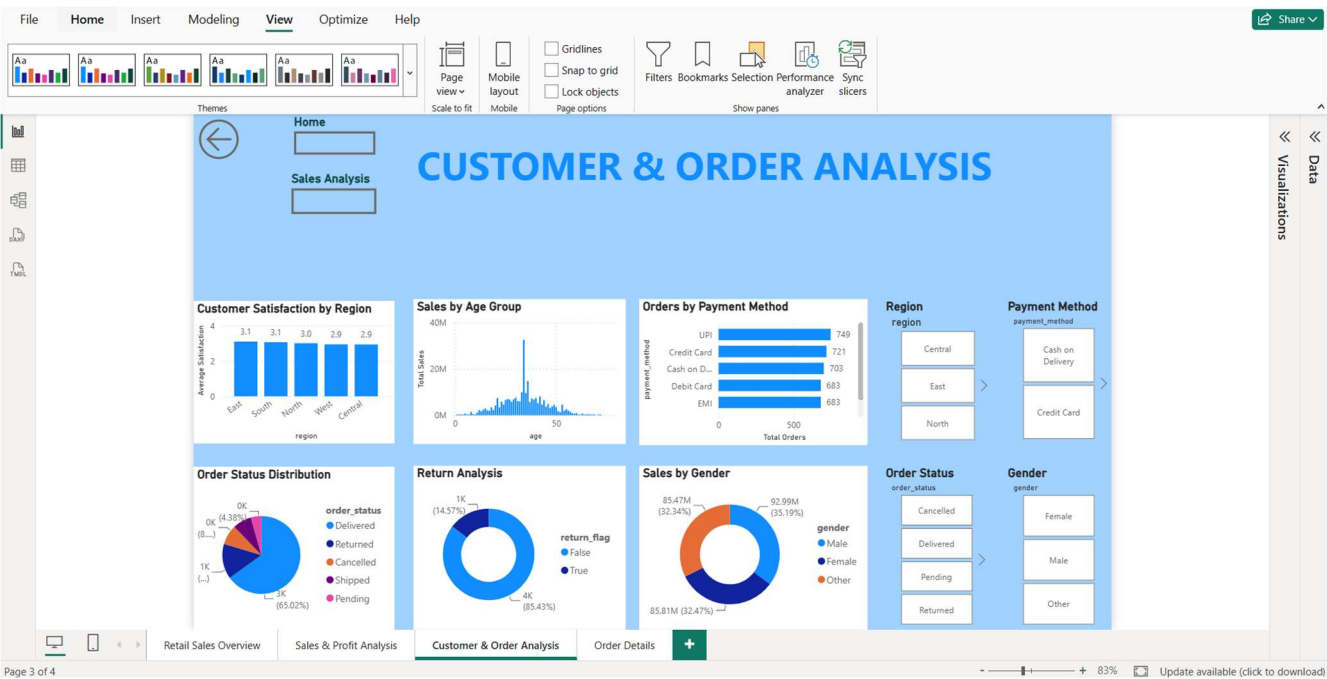
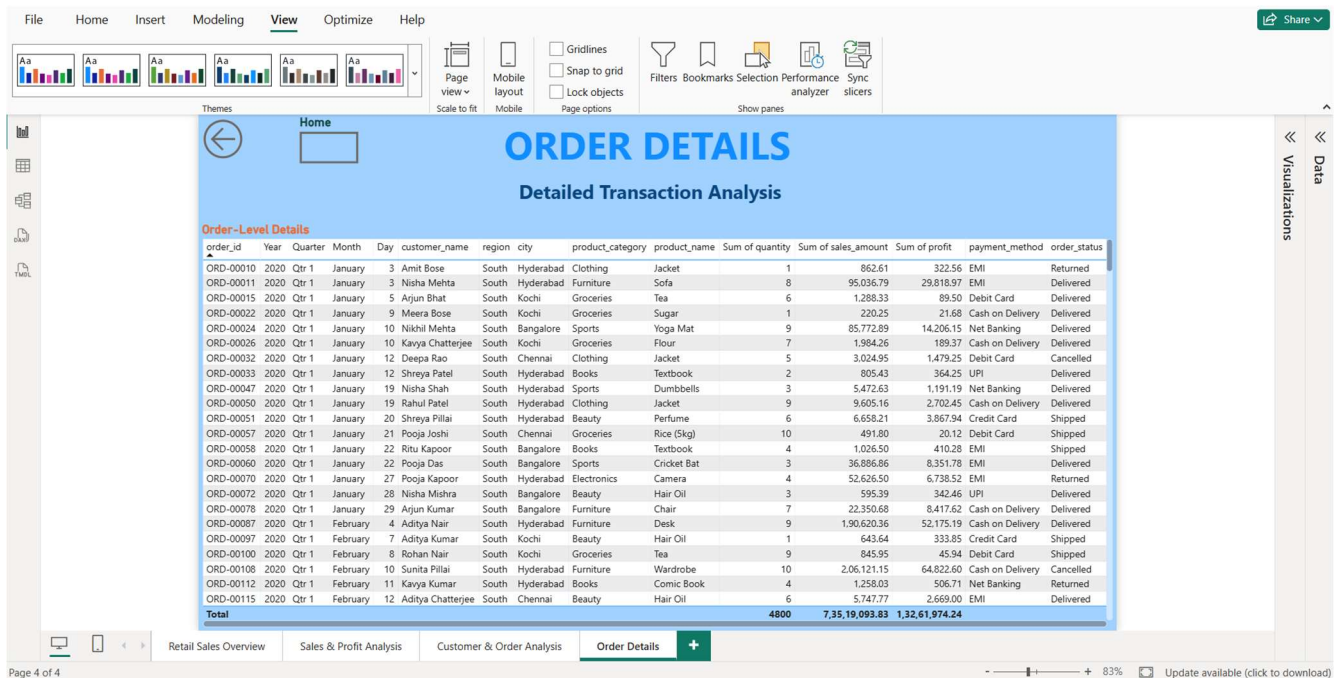


Figure 4 – Order Details



8. Key Insights / Findings

The dashboard provides a structured way to identify differences in sales, profit, customer behaviour and order patterns. The following findings are based on the completed dashboard structure and the visual analysis implemented in the Power BI report.

- Regional performance can be compared directly using Sales by Region and Total Profit by Region visuals.
- Product categories can be compared using both sales and profit measures, helping identify stronger and weaker categories.
- Monthly sales trend analysis makes it possible to observe changes in sales over the reporting period.
- Payment Method visuals show how orders are distributed across UPI, cards, EMI, Cash on Delivery and Net Banking.
- Customer analysis provides a view of satisfaction by region and sales distribution by gender.
- Order Status Distribution helps distinguish Delivered, Returned, Cancelled, Shipped and other order states.
- Return Analysis provides a direct view of returned versus non-returned transactions.
- Age-based sales analysis supports customer demographic exploration.
- Interactive slicers allow the same visuals to be examined for selected regions, categories, dates, genders, payment methods or order statuses.

- Region-based drill-through connects summary analysis with transaction-level details, improving the usefulness of the dashboard for investigation.

9. Conclusion

The Retail Sales Analysis mini project successfully combined Excel-based data preprocessing with Power BI-based interactive visualization. In Phase 1, the dataset was cleaned and validated through blank-row removal, missing-value handling, categorical standardization, date cleaning, numerical validation, duplicate checking, filtering and sorting. Calculated fields and Pivot Tables were also created.

In Phase 2, the cleaned data was converted into a four-page interactive Power BI dashboard. The report includes KPI cards, charts, slicers, navigation buttons, Sales View and Profit View bookmarks, and a region-based drill-through Order Details page. The result is a structured dashboard that makes retail sales, profit, customer and order analysis easier to explore and present.

10. Future Scope

- Forecasting: Future sales and demand forecasting can be added using Power BI forecasting or external machine-learning models.
- Real-time data: A live or frequently refreshed data source can be connected for near real-time monitoring.
- Automated reports: Scheduled refresh and automated report distribution can be implemented.
- Advanced KPIs: Additional profitability, customer retention and return-rate measures can be developed.
- Advanced analytics: Customer segmentation and product-level predictive analysis can be added.

11. References

The report is based primarily on the project files and the completed Excel/Power BI work used during the mini project.

- Retail Sales Mini Project – Phase 1: Excel Data Preprocessing and Analysis Report.
- Mini Project – Phase 2: Retail Sales Analysis Dashboard using Power BI.
- Microsoft Excel documentation and help resources used for spreadsheet data preparation.
- Microsoft Power BI documentation and help resources used for dashboard development and interactive features.

Dataset source: [Enter the original dataset source / website / provider used for the project].